

Technological trends in design

Since beginning of time, design has been a bridge between technology and its consumer. Hence design as a function always has to keep up with advancements in technology and it always did.

Talking of the current trends, Hari Nallan, experience strategist, co-founder and CEO, Think Design Collaborative, shares:

1. Currently, design innovation has gone social and this trend will define the products of tomorrow. Platforms like Kickstarter allow designers to make their own products and sell to public. Similarly, online platforms like Quirky allow social innovation. This is one trend that will stay.

2. In digital space, mobility is a trend that is catching up like rapid fire. What we understood as usability in the recent past is being challenged by advancements in this space.

3. Homes are becoming smart and we will be seeing rapid mass consumerisation of the smart homes concept. A middle-class user will be able to control his home from his mobile and this is already shaping up currently.

4. Lastly, technology and human interface (analogous to mechanism and product of the past) are rapidly converging. What this means to designers is that they can no more wrap a beautiful skin around technology and call it a good design. They both need to function in unison and that provides enough opportunities as well as challenges.

"IDHs are getting into complex designs in the areas of high-end consumer electronics (wearable gadgets), automotive infotainment, medical electronics and telecom. This involves designing complex hardware with blind and buried micro via, integration of multiple radios, using high-end processors (quad core) on a very small form factor with very good performance at reduced costs. Also, the current processors are very powerful with GHz+ processing power and multi-core architecture, requiring a high degree of proficiency in embedded software to take advantage of such high-end hardware platforms," informs Srinivas Panapakam, general manager-PES Sales and Business Development, Mistral Solutions.

"The trend is shaping towards bridging of multiple technologies. For instance, communication systems and information technology in the present age are too well spread, which enables user a greater experience. IDHs should constantly identify ways to integrate the two technologies in offering solutions to customers, which enables flexibility at multiple levels," adds Ganesh Shankar, co-founder and managing director of FluxGen Engineering Technologies.

The Internet of Things is another simple and impactful trend.

"With the advent of IoT (Internet of Things) and approximately 6.5 billion connected devices predicted by 2016, there would be a lot of advancement in all domains including vehicles, security, telemedicine and energy. There will be lots of opportunities like devices with human appeal, more intelligent machines, autonomous cars, smart towns and so on," explains R.K. Shenoy, senior vice president, Powertrain Electronics, Robert Bosch Engineering & Business Solutions.

"Connecting everything, everyone to the Internet will create huge opportunities. We are investing a lot in that area," adds Jithu Niruthambath, director, Calixto Systems.

Among other trends, "User-experience is gaining predominance over basic functionality and engineering. 3D printing could go mainstream along with virtual manufacturing," informs Shenoy.

capabilities and expertise in design development; and functioning as one-stop shop for design development needs of companies served."

"IDHs in India have to invariably compete with IDHs across countries, especially China. The catch is in identifying the niche and capturing the market where not many have entered that particular area. Sometime it is also about creating the niche which may not exist. The present-day technology advancement is very rapid. Continuously being updated about the indus-

try development is essential to make improvements in quality standards and perhaps developing features that add to the portfolio of the IDHs. Offering value-added service can also lead to a competitive advantage," says Ganesh Shankar, co-founder and managing director of FluxGen Engineering Technologies.

Niruthambath informs, "We are planning to grow by focusing on high-growth verticals, creating reference solutions to reduce customers' time to market and identifying similar markets."

"System/domain competence is a prerequisite for complete product development. This includes not only in application but software, hardware, mechanics and manufacturing as well. High investments are needed both in people and machines to achieve this. Long-term strategy with shared vision is key to success. Increased cooperation with universities to bridge the gap between the industry and academia is important, starting with having industry-relevant syllabus and lab setups. We are steadily increasing our efforts in this area and are also active with NASSCOM to strengthen this effort. Also, strong global managers and worldwide network of front-end technical sales team is becoming increasingly important because of the spread of our customers. This is currently our focus," says Shenoy.

Why not create own products

With all the expertise that IDHs have, why don't they create their own products, under their own brand name?

There are several reasons for this. "IDHs are primarily design houses that do product designs for an end customer, and the end customer owns the IP. So IDHs are more like 'designers for hire.' Also, if the IDHs start doing their own brand products, they cannot really work as a services entity. So the choice here is whether to stay as a services company or become a product company. The strength of the IDHs lies in their technical capability. Being a product company has other associated challenges like doing market survey (typically global), product marketing and branding, and product life cycle, which need significant investment. It is also tough to address the global market based in India. It is ideal to do that out of USA or other western geographies since you need to be closer to the customer," explains Panapakam.

Nallan has a similar answer. "Owning and operating a brand is a different ballgame altogether that requires high levels of motivation, will and, most